



Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • Transactional dataset



Data Prep & Cleaning

Load & Inspect

pandas: df.info(), .describe()

Missing Ratings

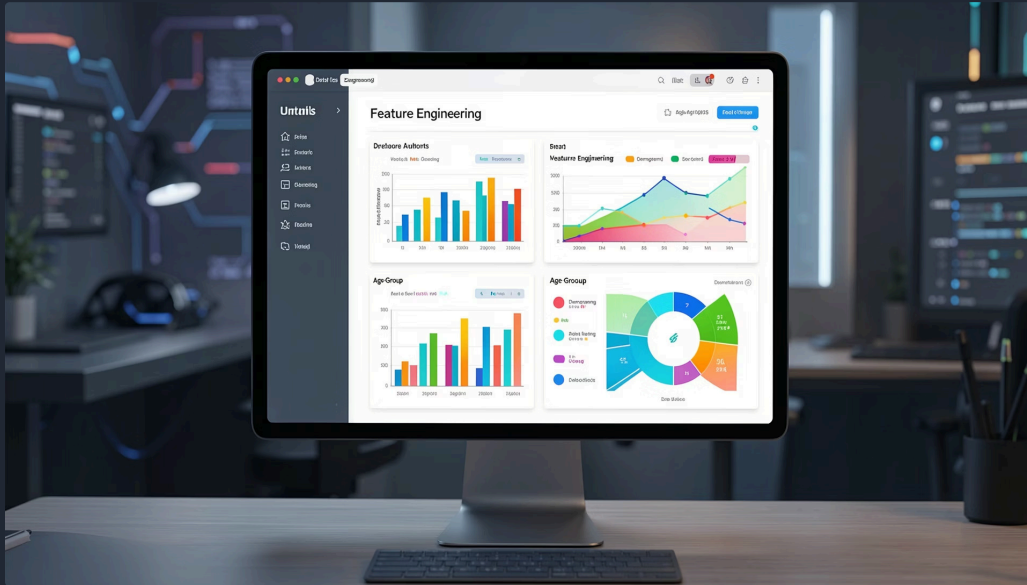
Imputed review rating by category median

Standardize

Renamed columns to snake case

DB Load

Loaded cleaned Data Frame to PostgreSQL



Feature Engineering

- age group from age bins
- purchase frequency days
- dropped promo code used (redundant with discount applied)

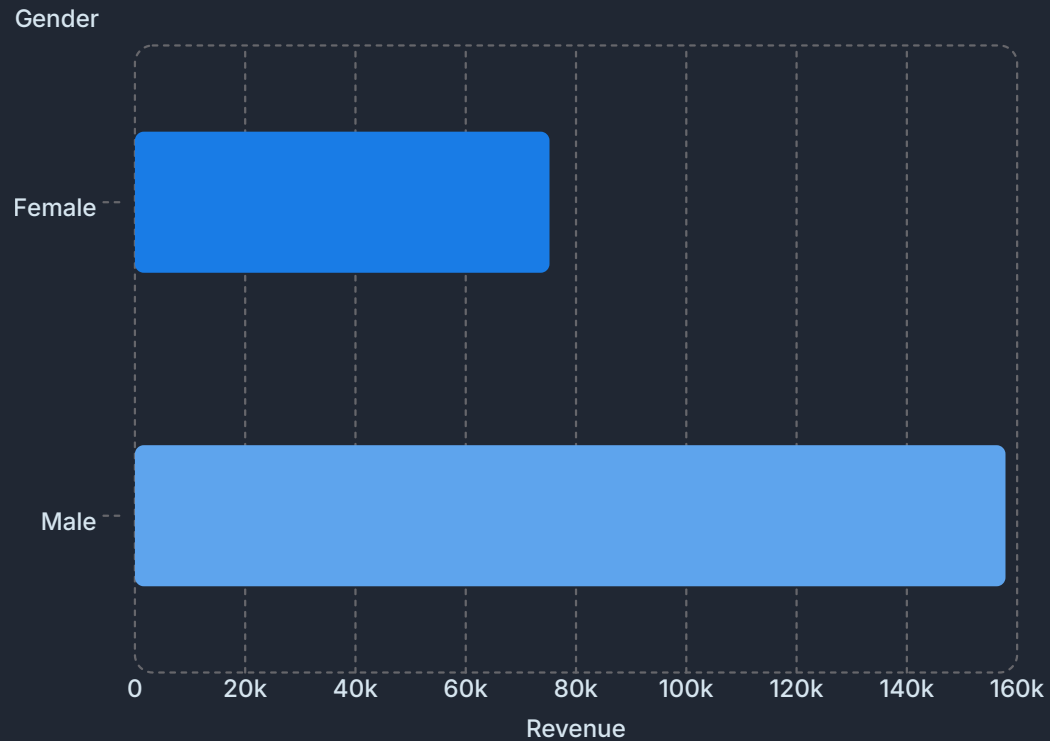
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SQL Analysis – Key Questions

Revenue by gender, high-spend discount users, top products, shipping comparison, subscribers vs non-subscribers

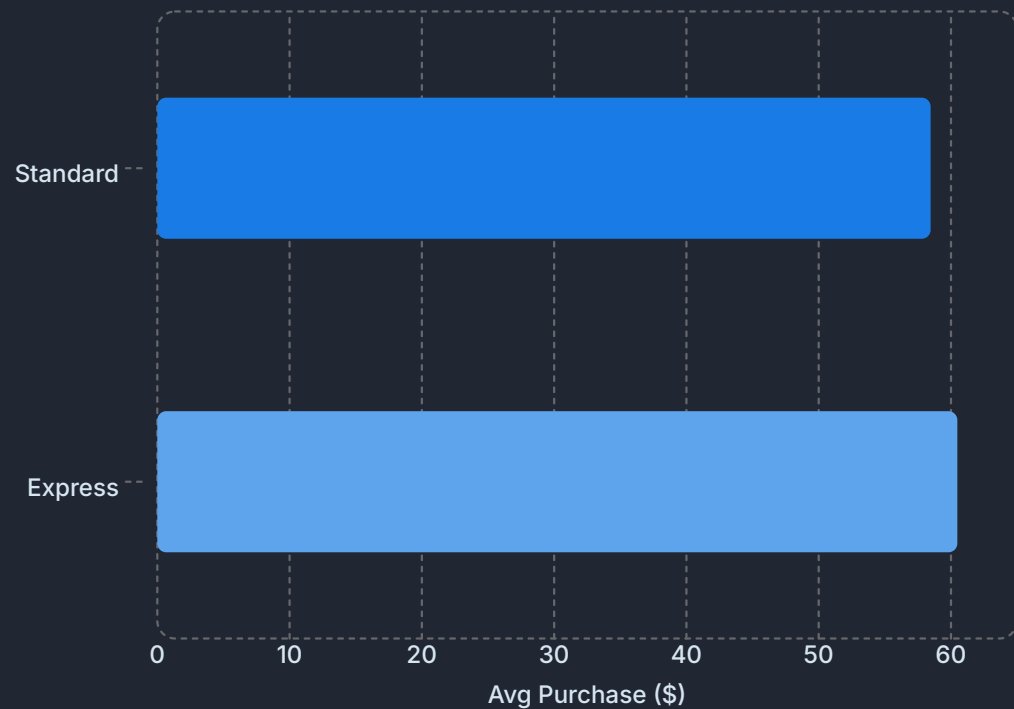


Revenue by Gender

Male > Female (157,890 vs 75,191)

📌 Inform targeting and assortment

Shipping Type



Shipping & Spend

Express slightly higher avg spend

Use shipping preferences for upsell



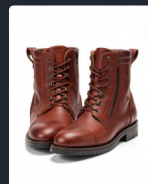
Gloves

Top rated — Avg
3.86



Sandals

Avg 3.84



Boots

Avg 3.82



Hat

Avg 3.80 — 50%
discount rate

Dashboard Highlights

- Customers: 3.9K
- Avg purchase: \$59.76
- Avg review: 3.75
- Subscribers: 27% (Yes) • 73% (No)

📄 Power BI used for interactive visuals





Segmentation & Loyalty

Loyal

3,116 customers

Returning

701 customers

New

83 customers

Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers to increase loyalty

Discount Strategy

Review discounts to protect margins

Product Positioning

Feature top-rated & best-selling items

Targeted Marketing

Focus on high-revenue age groups & express shippers